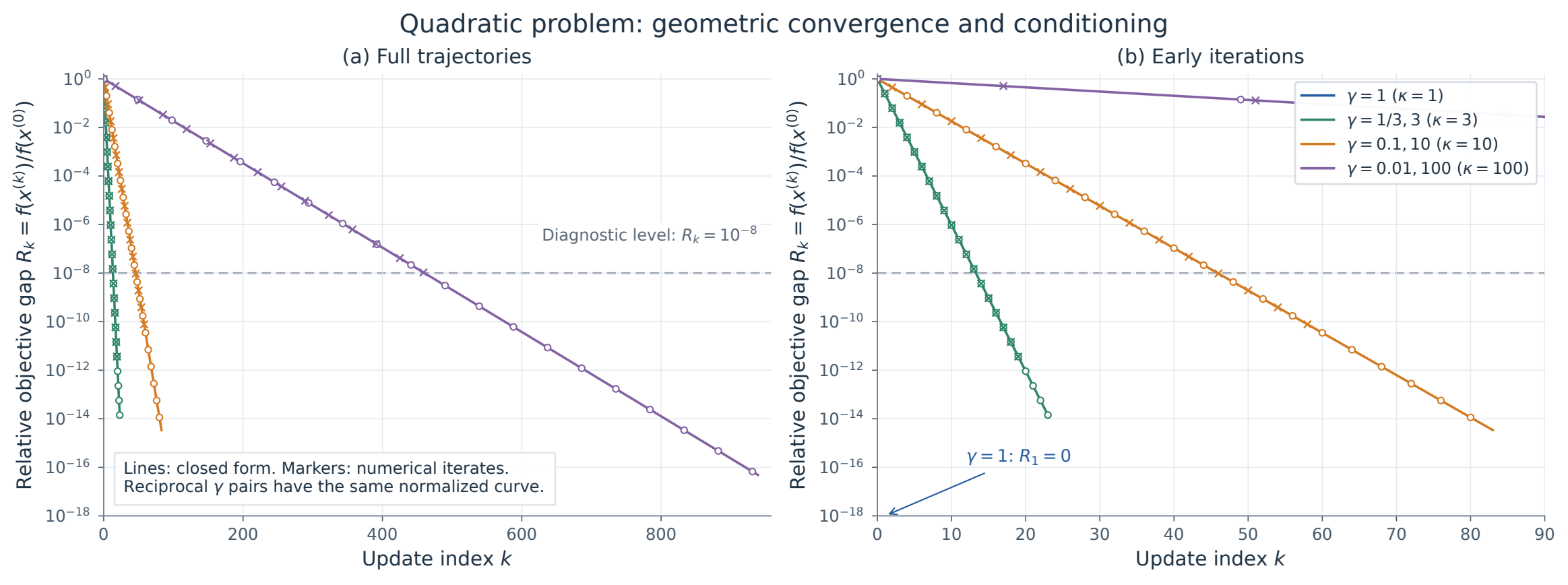
Quadratic problem: geometric convergence and conditioning
(a) Full trajectories
(b) Early iterations
Relative objective gap $R_k = f(x^{(k)})/f(x^{(0)})$
Update index k
Diagnostic level: $R_k = 10^{-8}$
Lines: closed form. Markers: numerical iterates.
Reciprocal γ pairs have the same normalized curve.
$\gamma = 1$ ($\kappa = 1$)
$\gamma = 1/3, 3$ ($\kappa = 3$)
$\gamma = 0.1, 10$ ($\kappa = 10$)
$\gamma = 0.01, 100$ ($\kappa = 100$)
$\gamma = 1$: $R_1 = 0$